Submitted to: Smart Internz , Smart bridge Company

Track : Data Analytics with Tableau Project

title : Heart Disease analysis Submitted by:

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1.Executive summary

1.1.Purpose

This project aims to analyze heart disease prevalence and its correlation with various demographic, lifestyle, and health-related factors using Tableau as the primary data visualization tool. The analysis provides actionable insights into the key risk factors contributing to heart disease, enabling healthcare stakeholders to understand patterns and make data-driven decisions.

1.2 Key Deliverables

- **Three Interactive Dashboards:**
- Demographic Analysis
- Lifestyle and Risk Factors
- Overall Analysis
- **Data Stories** for executive presentations
- **Interactive Filters** for drill-down analysis
- **Key Insights Report** summarizing findings

1.3 Summary of Findings

Finding	Insight
Age Factor	Heart disease prevalence increases significantly with age
Gender	Males show higher susceptibility to heart disease than females
BMI Impact	Higher BMI correlates strongly with heart disease incidence
Lifestyle	Smoking, alcohol consumption, and physical inactivity are major risk factors
Comorbidities	Diabetes, stroke, kidney disease, and asthma increase heart disease risk
Health Status	Poor general health and limited physical activity are strongly associated

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2. INTRODUCTION & BACKGROUND

2.1 Problem Statement

leading cause of death

Heart disease remains th eaths annually. Early de globally,
17 million understanding o nd accounting for
crucial for:

- Reducing mortality rates
- Implementing preventive healthcare measures
- Optimizing resource allocation in healthcare systems

2.2 Need for Analytics

Traditional reporting often fails to uncover hidden patterns and correlations within complex healthcare data. This project leverages **data analytics** and **visualization** to:

1. Identify high-risk populations
2. Understand the relationship between lifestyle factors and heart disease
3. Provide interactive tools for healthcare decision-makers

2.3 Project Scope

Scope Area	Description
Data Scope	Comprehensive health survey data with 15+ variables
Geographic Scope	National/Regional healthcare data
Time Scope	Cross-sectional analysis
Analytical Scope	Descriptive & diagnostic analytics

PROJECT 3. OBJECTIVES

3.1 Primary Objectives

	Objective	Success Criteria
1	Demographic Analysis	Analyze heart disease prevalence across age, gender, and race categorie

	Objective	Success Criteria
2	Lifestyle Factor Analysis	Identify correlation between lifestyle choices (smoking, alcohol, physic

3	Health Comorbidity Analysis disea	Examine relationship between existing health conditions and heart
4	Interactive Visualization	Create user-friendly dashboards enabling drill-down analysis 5 Insights
	Generation	Derive actionable recommendations for healthcare providers

3.2 Secondary Objectives

- Establish a reusable analytics framework for future healthcare projects
- Demonstrate proficiency in Tableau visualization techniques
- Publish findings through Tableau Stories for stakeholder presentations

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4. DATA DESCRIPTION

4.1 Data Source

Kaggle

4.2 Data Dictionary

Variable Name	Data Type	Possible Values
Heart Disease	Boolean/Categorical	Yes / No
BMI	Continuous	10 - 60+
Smoking	Categorical	Current/Former/Never
Alcohol Drinking	Categorical	Heavy/Moderate/None
Stroke	Boolean	Yes / No
Physical Health	Discrete	0-30
Mental Health	Discrete	0-30
Diff Walking	Boolean	Yes / No
Sex	Categorical	Male / Female
Age Category	Categorical	18-24, 25-29, etc.
Race	Categorical	White, Black, Asian, Hispanic, etc.
Diabetic	Categorical	Yes / No / Pregnancy
Variable Name	**Data Type**	**Possible Values**
Physical Activity	Categorical	Active / Inactive
Gen Health	Categorical	Excellent/Poor

Sleep Time	Continuous	0-24
Asthma	Boolean	Yes / No
Kidney Disease	Boolean	Yes / No
Skin Cancer	Boolean	Yes / No

4.3 Dataset Statistics

Metric	**Value**
Total Records	~400,000+
Variables	18
Data Quality	Validated

5. METHODOLOGY

5.1 Data Processing Pipeline

5.1 Data Processing Pipeline

1. Data Collection
2. Data Cleaning (handle missing values, outliers)
3. Data Transformation (create derived fields)
4. Tableau Import
5. Visualization Creation
6. Dashboard Building
7. Story Creation

5.2 Data Cleaning Steps

- Missing Value Treatment
- Outlier Detection
- Data Type Conversion
- Derived Variables Creation (Age Groups)

6. DASHBOARD DESCRIPTIONS

Purpose

To understand how heart disease prevalence varies across different demographic characteristics.

Key Visualizations

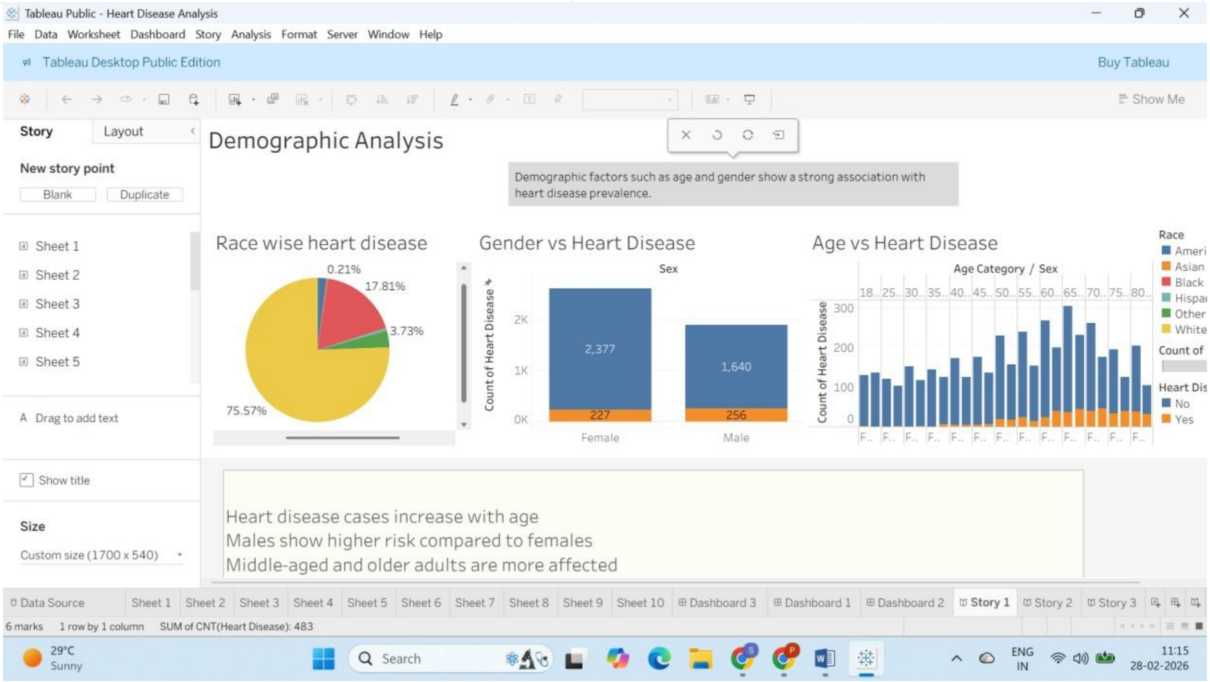
	Visualization	Chart Type
1	Age Distribution	Histogram/Bars
2	Gender Analysis	Side-by-side Bars
3	Racial Distribution	Stacked Bar/Pie
4	Age vs Heart Disease	Line Chart
5	Demographic Summary	KPI Cards

Filters Available

- Age Category
- Sex
- Race

Key Insights

- Heart disease prevalence increases with age
- Males have higher risk than females



6.2 Dashboard 2: Lifestyle & Risk Factors

Purpose

To analyze the impact of lifestyle choices and behavioral factors on heart disease incidence.

Key Visualizations

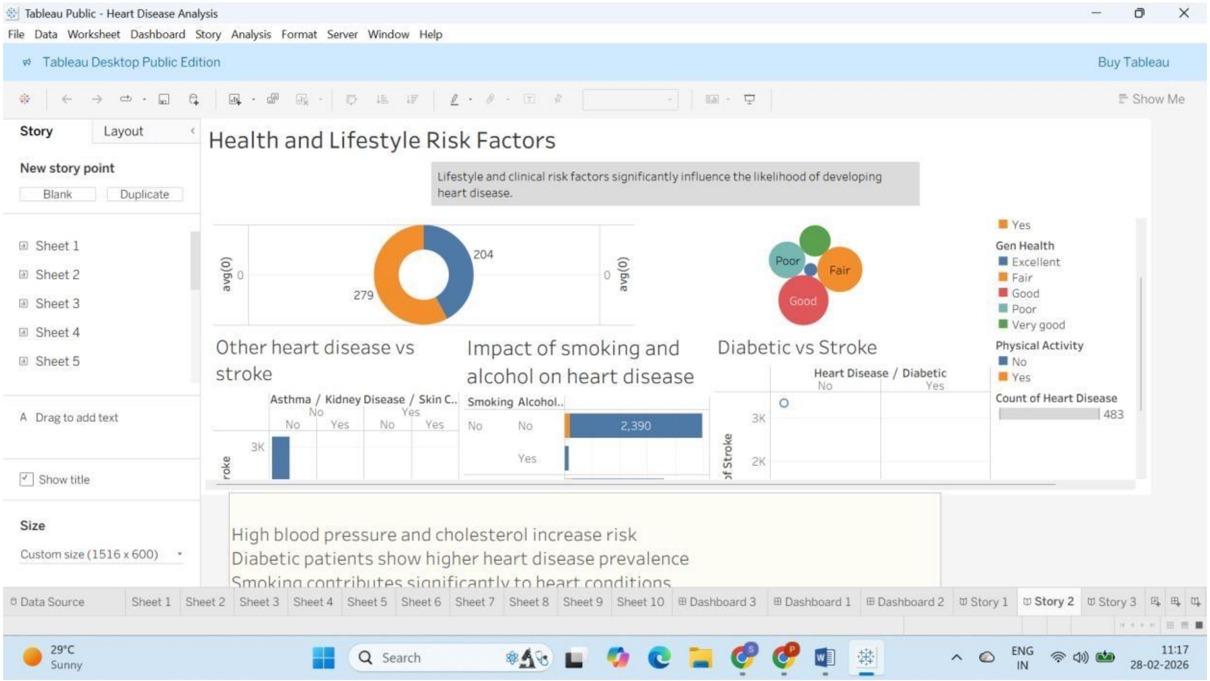
	Visualization	Chart Type
1	Smoking Analysis	Grouped Bar
2	Alcohol Consumption	Stacked Bar
3	BMI Distribution	Histogram
4	Physical Activity	Side-by-side Bars
5	Sleep Patterns	Box Plot
6	Health Conditions	Heat Map

Filters Available

- Smoking Status
- Alcohol Level
- BMI Range
- Physical Activity Level
- Sleep Time Range

Key Insights

- Smoking significantly increases heart disease risk
- Physical inactivity is a major contributor
- Obesity strongly associated with heart disease



6.3 Dashboard 3: Overall Analysis

Purpose

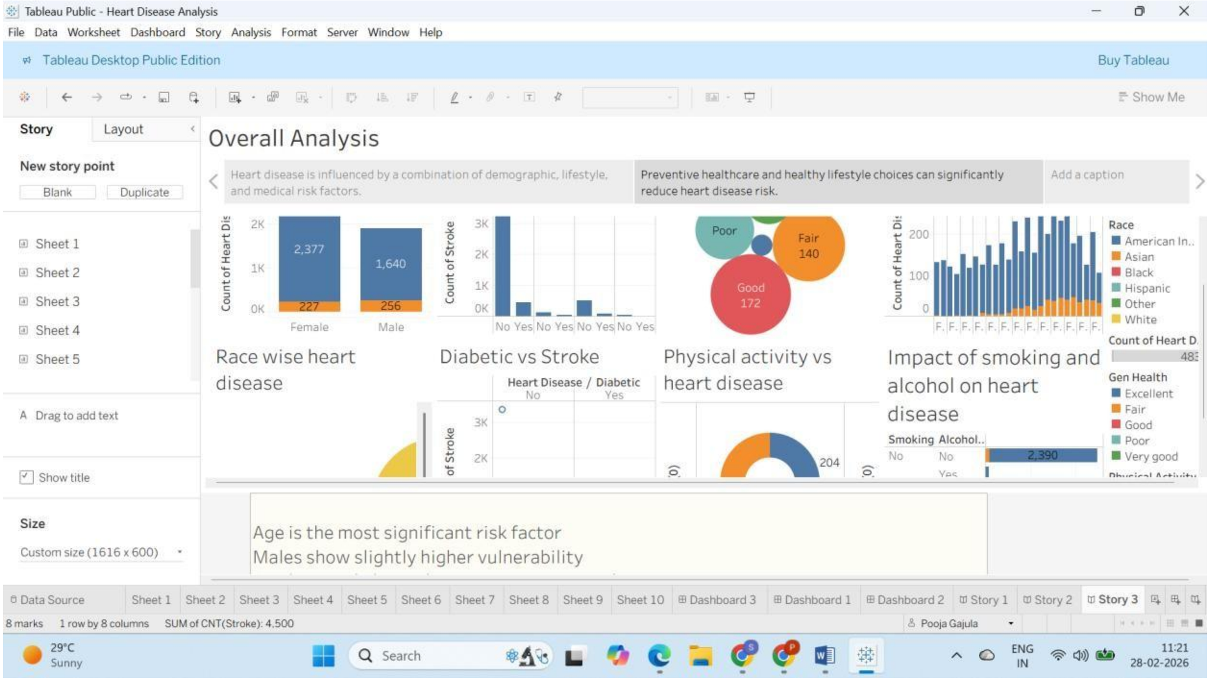
To provide a comprehensive overview combining all factors and presenting high-level findings.

Key Visualizations

	Visualization	Chart Type
1	Heart Disease Overview	KPI Dashboard
2	Risk Factor Ranking	Horizontal Bar
3	Comorbidity Analysis	Stacked Bar
4	Health Status Impact	Diverging Bar
5	Summary Statistics	Table

Filters Available

- All combined filters



7. KEY INSIGHTS & FINDINGS

7.1 Demographic Insights

Finding	Description
Age Correlation	Prevalence increases after age 45
Gender Disparity	Males show ~20% higher rate
Racial Variations	Some groups show 2-3x higher risk

7.2 Lifestyle Insights

Factor	Impact
Smoking	High - 2-3x higher risk
Physical Inactivity	High - 40% higher prevalence
BMI	High - Strongly correlated
Alcohol	☐ Moderate
Sleep	☐ Moderate

7.3 Health Comorbidity Insights

Condition	Correlation
Diabetes	Strong positive
Stroke	High co-occurrence
Kidney Disease	Moderate
Asthma	Slight increase

8.1 Primary Tools

Tool	Purpose	Version
Tableau Desktop	Visualization & Dashboards	2024+
Tableau Public	Publishing & Sharing	Latest

8.2 Supporting Tools

Tool	Purpose
Microsoft Excel	Initial data exploration
Mysql	Data cleaning
PowerPoint	Presentation

8.3 Tableau Features Used

- ☑ Calculated Fields
- ☑ Parameters & Filters
- ☑ Sets and Groups
- ☑ Dual Axis Charts
- ☑ Storytelling
- ☑ Dashboard Actions
- ☑ Tooltips Customization
- ☑ KPI Indicators

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9. PROJECT ARCHITECTURE

9.1 High-Level Architecture

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│ DATA SOURCE │

(BRFSS Survey Data / CSV/Excel)

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DATA PREPARATION

(Excel / Tableau Prep / Python Cleaning)

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TABLEAU WORKBOOK

Demographic Analysis Dashboard	Lifestyle & Risk Factors Dashboard	Overall Analysis Dashboard

DATA STORIES

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PUBLICATION & SHARING

| (Tableau Public / Tableau Server / Tableau Online) |

11. CONCLUSIONS & RECOMMENDATIONS

11.1 Conclusions

1. **Heart disease is multifactorial**: Combination of demographic, lifestyle, and health factors
2. **Age is strongest predictor**: Prevalence increases after age 45
3. **Lifestyle modifications can reduce risk**: Smoking cessation, exercise, healthy BMI
4. **Comorbidities amplify risk**: Diabetes, stroke, kidney disease require monitoring
5. **Data-driven insights enable action**: Interactive dashboards empower stakeholders

11.2 Recommendations

Area	Recommendation
Screening	Target 45+ age group
Prevention	Promote smoking cessation programs
Healthcare	Integrate BMI monitoring in primary care
Policy	Develop lifestyle intervention programs
Future	Add predictive modeling for risk scoring

Appendix A: Data Field Mapping

Original Field	Tableau Field Name	Data Type
Heart Disease	Heart_Disease	Boolean
BMI	BMI	Number (Decimal)
Original Field	**Tableau Field Name**	**Data Type**

Smoking	Smoking_Status	String
Alcohol Drinking	Alcohol_Consumption	String
Stroke	Stroke_History	Boolean
Physical Health	Physical_Health_Days	Integer
Mental Health	Mental_Health_Days	Integer
Diff Walking	Difficulty_Walking	Boolean
Sex	Gender	String
Age Category	Age_Group	String
Race	Race	String
Diabetic	Diabetes_Status	String
Physical Activity	Physical_Activity	String
Gen Health	General_Health	String
Sleep Time	Sleep_Hours	Number (Decimal)
Asthma	Asthma_History	Boolean
Kidney Disease	Kidney_Disease_History	Boolean
Skin Cancer	Skin_Cancer_History	Boolean

Appendix B: Dashboard Filters Configuration

Dashboard	Filter	Type	Source Field
All	Age Category	Wildcard	Age_Group
All	Sex	Single Value	Gender
All	Race	Multi-Select	Race
Lifestyle	Smoking Status	Multi-Select	Smoking_Status
Lifestyle	BMI Range	Range	BMI
Lifestyle	Physical Activity	Single Value	Physical_Activity

Dataset link:

https://drive.google.com/file/d/190Qmq27LeZZ_nWricP3Obl7ys_5otEsp/view

Tableau public link:

https://public.tableau.com/views/HeartDiseaseAnalysis_17721260095200/Story3?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link